# SeRP Linkage Report

## Splink Cohort Builder

Linkage Project Name: splink_cohort_builder

Linkage Run Name: Run 1

Linkage Type: Link Dedupe

Model Type: Probabilistic

Document Creation Date: 2026-07-07 09:31

### Run Configuration

| | |
|---|---|
| **Operation mode:** | Link Dedupe |
| **Linkage type:** | Probabilistic |
| **Fields used:** | first_name, surname, dob, city, email, gender, postcode |
| **Active blocking:** | first_name, surname, dob, city, email, gender, postcode |
| **Cluster threshold:** | 0.8 |

# Datasets

This section shows an overview of the datasets used, including linkage fields, field mapping, and column-level completeness (missingness).

### Base table name: Dataset A (fake1000a)

**Dataset:** Splink fake_1000 + gender (name inference) + postcode (UK GeoNames)

**Number of rows:** 1000

### Linkage fields

first_name, surname, dob, city, email, gender, postcode

### Field mapping (Dataset A)

| Linkage System Alias | Source Fieldname |
|---|---|
| first_name | first_name |
| surname | surname |
| dob | dob |
| city | city |
| email | email |
| gender | gender |
| postcode | postcode |

### Dataset statistics: Dataset A

**Missingness:** Completeness per column shown below.

| Field | Completeness (%) |
|---|---|
| first_name | 83.1% |
| surname | 81.9% |
| dob | 100.0% |
| city | 81.3% |
| email | 78.9% |
| gender | 100.0% |
| postcode | 57.7% |

## Base table name: Dataset B (fake1000b)

Dataset B is a 50% sample of Dataset A with controlled data quality errors: 14% first-name typos, 9% surname typos, 5% missing DOBs, 15% email variations, 11% city abbreviations, 7% gender errors.

**Number of rows:** 500

## Field mapping (Dataset B)

| Linkage System Alias | Source Fieldname |
|---|---|
| first_name | first_name |
| surname | surname |
| dob | dob |
| city | city |
| email | email |
| gender | gender |
| postcode | postcode |

## Dataset statistics: Dataset B

| Field | Completeness (%) |
|---|---|
| first_name | 83.2% |
| surname | 82.6% |
| dob | 94.2% |
| city | 80.8% |
| email | 77.4% |
| gender | 100.0% |
| postcode | 58.4% |

## Column Completeness Chart

Completeness = percentage of non-null values per column.

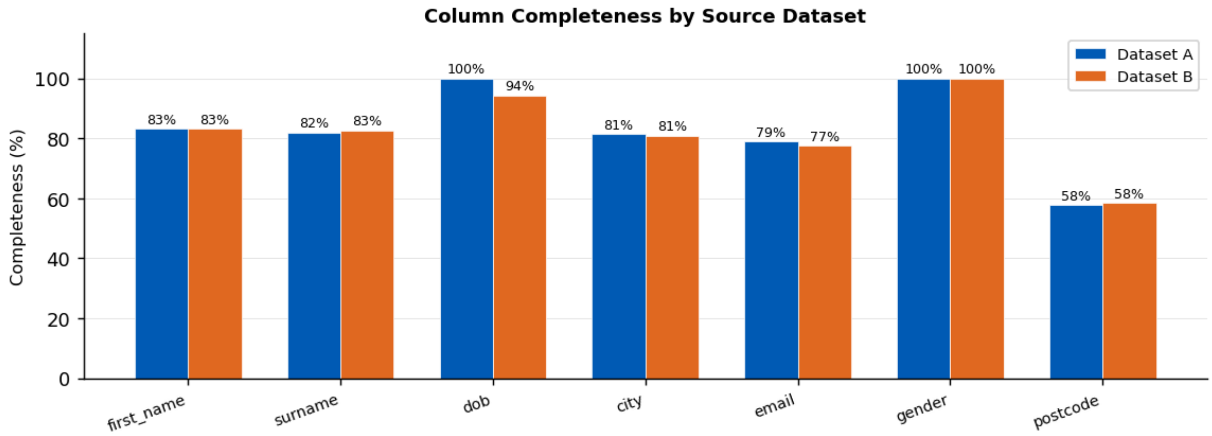


*Figure: Column completeness by source dataset.*

# Blocking Rules

Blocking rules generate pairwise record comparisons. Only record pairs that agree on at least one blocking field are compared. Blocking reduces the O(n^2) comparison space but may miss true matches if rules are too restrictive.

### Blocking rule 0:

```
l."first_name" = r."first_name"
```

**Number of generated comparisons:** 997

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

### Blocking rule 1:

```
l."surname" = r."surname"
```

**Number of generated comparisons:** 538

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

### Blocking rule 2:

```
l."dob" = r."dob"
```

**Number of generated comparisons:** 339

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

### Blocking rule 3:

```
l."city" = r."city"
```

**Number of generated comparisons:** 277

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

### Blocking rule 4:

```
l."email" = r."email"
```

**Number of generated comparisons:** 75

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

### Blocking rule 5:

```
l."gender" = r."gender"
```

**Number of generated comparisons:** 259

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

### Blocking rule 6:

```
l."postcode" = r."postcode"
```

**Number of generated comparisons:** N/A

Reasoning: Pairs that agree on this field are brought forward for detailed comparison. Useful when this field has high completeness and low error rate.

## Cumulative count of blocking rules chart

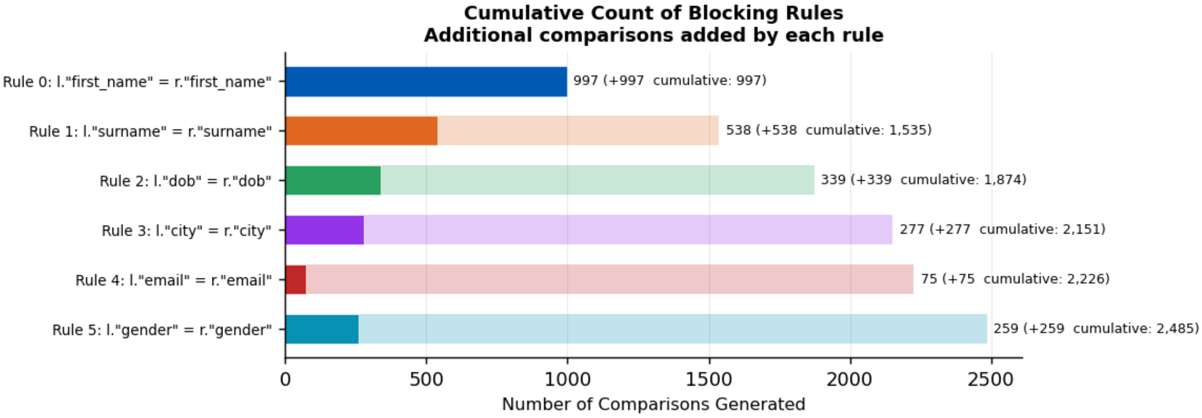


*Figure: Additional comparisons generated by each successive blocking rule.*

# Comparison Methods

Comparison methods are used to score the relationship between two records. Each method defines ordered levels from exact match to total disagreement. A null level handles missing values with no evidence contribution.

**Number of comparison methods for edge scoring: 7**

### Comparison 0: first_name

SQL comparison levels (first match wins):

```
"first_name_l" IS NULL OR "first_name_r" IS NULL  [NULL LEVEL]
"first_name_l" = "first_name_r"
jaro_winkler_similarity("first_name_l", "first_name_r") >= 0.92
jaro_winkler_similarity("first_name_l", "first_name_r") >= 0.88
jaro_winkler_similarity("first_name_l", "first_name_r") >= 0.7
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

### Comparison 1: surname

SQL comparison levels (first match wins):

```
"surname_l" IS NULL OR "surname_r" IS NULL  [NULL LEVEL]
"surname_l" = "surname_r"
jaro_winkler_similarity("surname_l", "surname_r") >= 0.92
jaro_winkler_similarity("surname_l", "surname_r") >= 0.88
jaro_winkler_similarity("surname_l", "surname_r") >= 0.7
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

### Comparison 2: dob

SQL comparison levels (first match wins):

```
try_strptime("dob_l", '%Y-%m-%d') IS NULL OR try_strptime("dob_r", '%Y-%m-%d') IS NULL  [NULL LEVEL]
"dob_l" = "dob_r"
damerau_levenshtein("dob_l", "dob_r") <= 1
ABS(EPOCH(try_strptime("dob_l", '%Y-%m-%d')) - EPOCH(try_strptime("dob_r", '%Y-%m-%d'))) <= 2629800.0
ABS(EPOCH(try_strptime("dob_l", '%Y-%m-%d')) - EPOCH(try_strptime("dob_r", '%Y-%m-%d'))) <= 31557600.0
ABS(EPOCH(try_strptime("dob_l", '%Y-%m-%d')) - EPOCH(try_strptime("dob_r", '%Y-%m-%d'))) <= 315576000.0
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

### Comparison 3: city

SQL comparison levels (first match wins):

```
"city_l" IS NULL OR "city_r" IS NULL  [NULL LEVEL]
```

```
"city_l" = "city_r"
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

### Comparison 4: email

SQL comparison levels (first match wins):

```
"email_l" IS NULL OR "email_r" IS NULL  [NULL LEVEL]
```

```
"email_l" = "email_r"
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

### Comparison 5: gender

SQL comparison levels (first match wins):

```
"gender_l" IS NULL OR "gender_r" IS NULL  [NULL LEVEL]
```

```
"gender_l" = "gender_r"
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

### Comparison 6: postcode

SQL comparison levels (first match wins):

```
"postcode_l" IS NULL OR "postcode_r" IS NULL  [NULL LEVEL]
```

```
"postcode_l" = "postcode_r"
```

ELSE (all other comparisons)

Description: CustomComparison
Reasoning: Placeholder text.

# Model Training

This section shows the model training parameters and hyperparameters used for the probabilistic Fellegi-Sunter model.

## Overview

**Number of training schemes:** 2 (prior estimate + EM)

## Prior Estimation

**Number of random samples:** 1.00E+05

**Recall estimate:** 0.6

Rule 0 (prior estimation):

```
l."first_name" = r."first_name"
```

## Estimation Schemes (Expectation-Maximisation)

fix_u_probabilities = True: u-probabilities held fixed during EM.

### Training Rule 0:

```
l."first_name" = r."first_name"
```

### Training Rule 1:

```
l."surname" = r."surname"
```

## Unlinkable Records

Some records lack sufficient information to produce a match weight above a given threshold. This chart shows what percentage will be unlinkable at each threshold, guiding the choice of operating threshold.

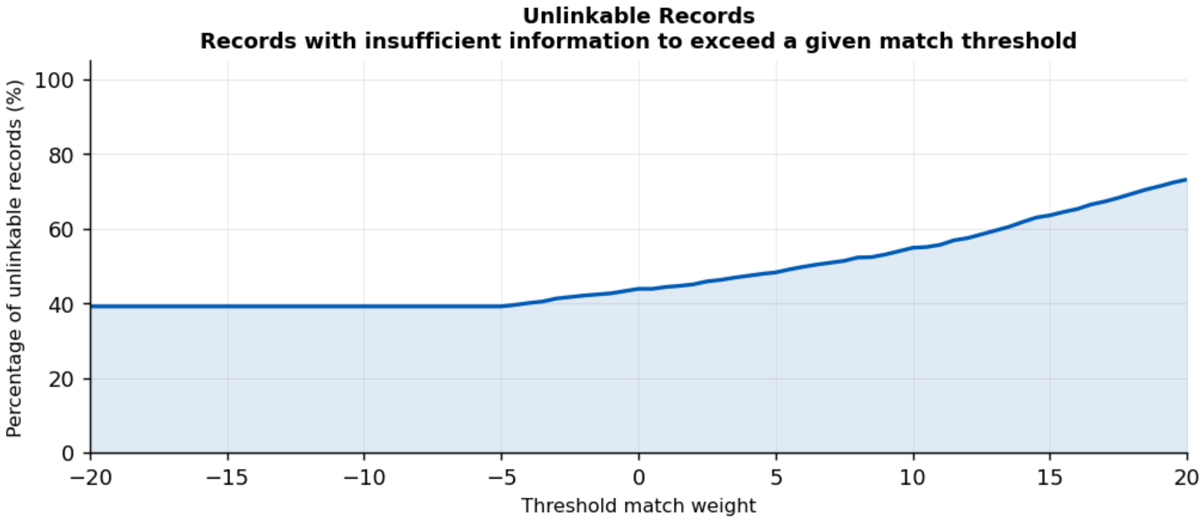


*Figure: Percentage of unlinkable records vs match-weight threshold.*

# Edge Metrics

This section shows the results of model inference of pairwise edges. Quality review insights are provided for the number of edges predicted, distribution of match weights, and match probabilities.

### Edge metrics overview:

**Number of predicted edges:** 2,485

**Number of distinct unique_ids with an edge:** 1,412

### Match probability statistics

| Statistic | Value |
|---|---|
| Mean | 0.742 |
| Median | 0.9935 |
| Min | 0.0303 |
| Max | 1.0 |
| Std Dev | 0.3751 |

### Match Weight Histogram

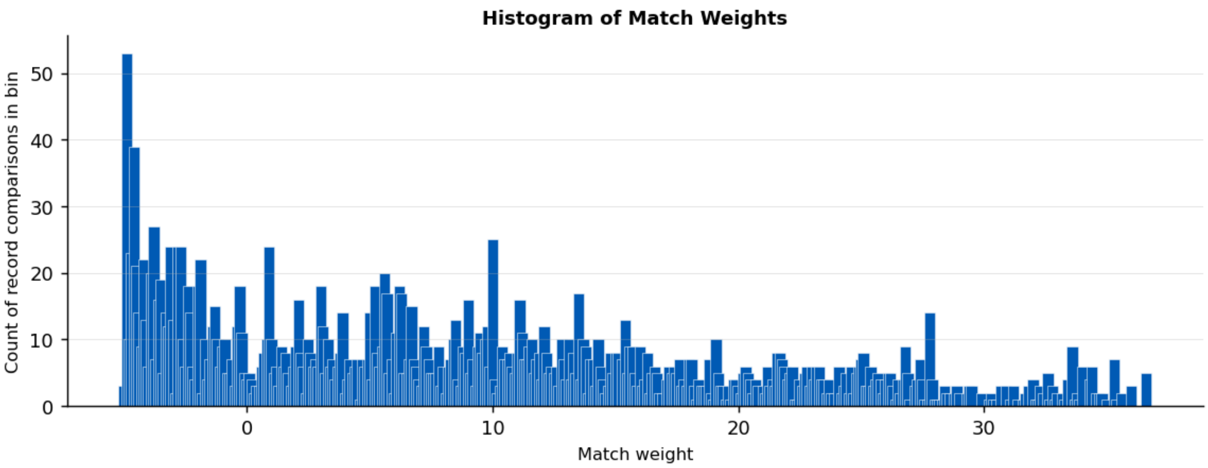


*Figure: Histogram of match weights. Peaks near high positive values indicate confident predictions.*

### Mean gamma scores by field

| Field | Mean Gamma |
|---|---|
| first_name | 1.9513 |
| surname | 1.7948 |
| dob | 3.0463 |
| city | 0.1404 |
| email | -0.0414 |
| gender | 0.6954 |
| postcode | -0.3525 |

# Cluster Metrics

This section shows the cluster inference results. A threshold was applied to the predicted edges to form entity clusters using a connected-components algorithm. Each cluster ideally represents one real-world individual.

### Cluster metrics overview:

**Number of clusters:** 440

**Cross-dataset clusters:** 262

| cluster_type | n_clusters | total_records |
|---|---|---|
| Multi-record cluster (2+ records) | 262 | 1322.0 |
| Singleton (1 record) | 178 | 178.0 |

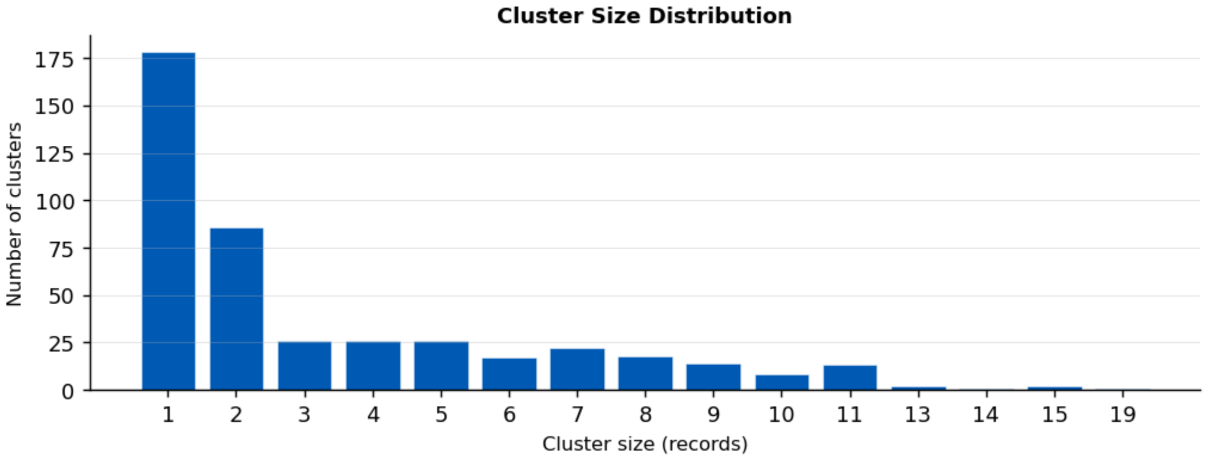


*Figure: Cluster size distribution.*

## Dataset overlap Venn diagram

The Venn diagram shows how many clusters contain records from Dataset A only, Dataset B only, or both datasets. Cross-dataset clusters represent successfully linked records.

| Category | N clusters |
|---|---|
| A only | 177 |
| B only | 1 |
| Both A and B | 262 |

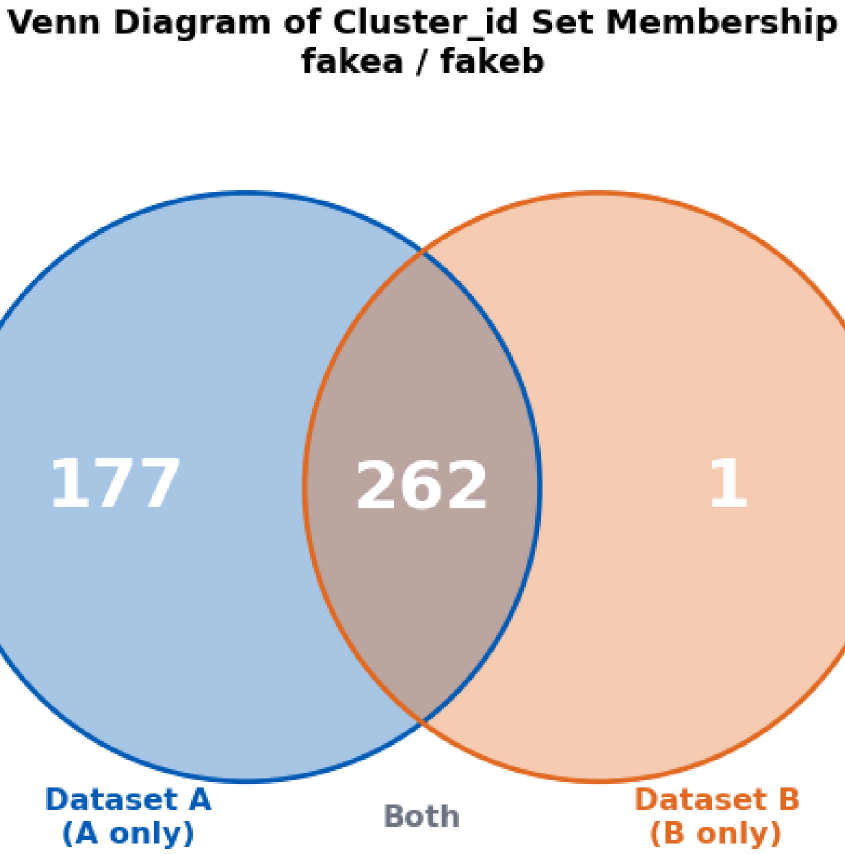


*Figure: Venn diagram of cluster_id set membership.*

## Model Accuracy: Confusion Matrix

Ground truth: the 'cluster' column in the input datasets. Records with the same cluster value are true matches. TP = predicted match AND true match. FP = predicted match BUT not true match. FN = true match NOT predicted. TN is omitted (too large for pairwise comparison).

**True Positives (TP):** 2,025
**False Positives (FP):** 460
**False Negatives (FN):** 528
**Ground truth pairs:** 2,553
**Predicted pairs:** 2,485

### Derived Metrics

| Metric | Value | Interpretation |
|---|---|---|
| Precision | 0.8149 | TP / (TP+FP) - of predicted matches, how many are correct |
| Recall | 0.7932 | TP / (TP+FN) - of true matches, how many were found |
| F1 Score | 0.8039 | Harmonic mean of Precision and Recall |
| F* Score | 0.6721 | TP / (TP+FP+FN) - accounts for both error types |
| FDR | 0.1851 | False Discovery Rate: FP / (TP+FP) |
| FNR | 0.2068 | False Negative Rate: FN / (TP+FN) |

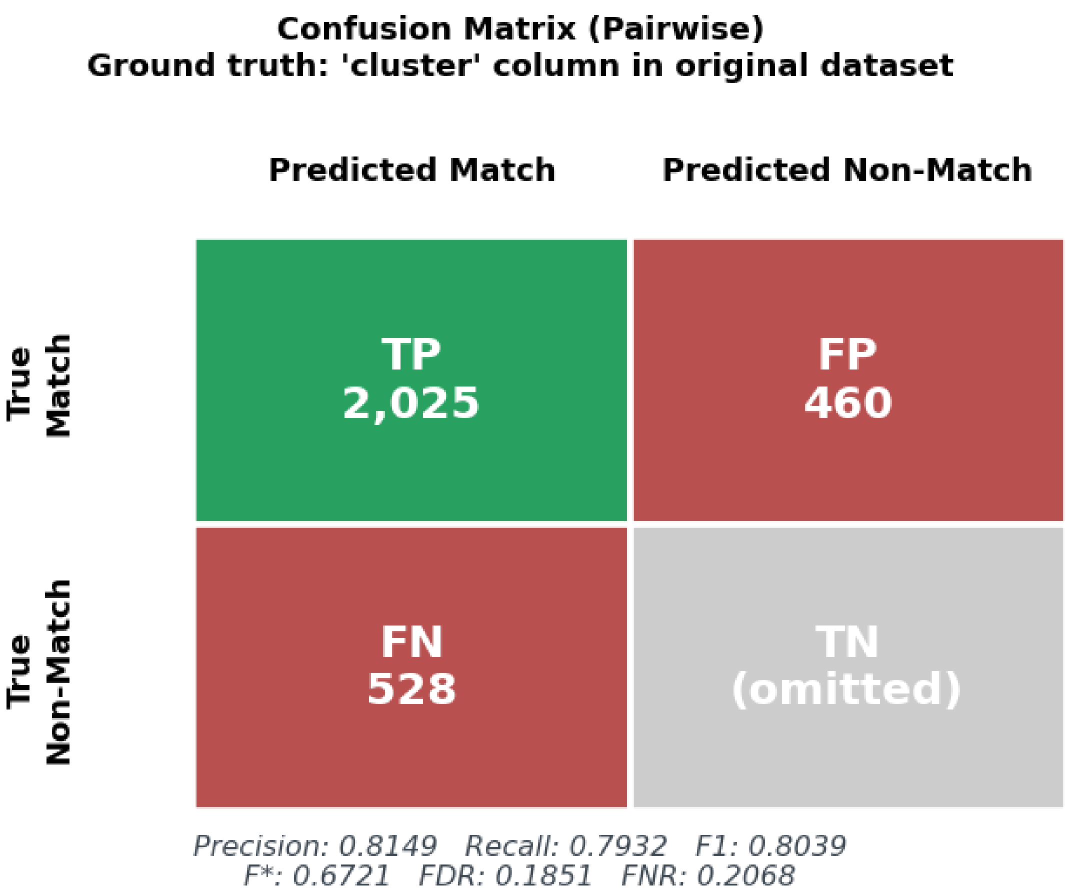


*Figure: Confusion matrix (pairwise). TN omitted as it is too large to count practically.*

## Precision-Recall Curve and F* Score (example 16 - CRL analysis)

These curves show Precision and Recall at every match-probability threshold. The F* curve shows the composite score TP/(TP+FP+FN). The ideal operating point maximises both Precision and Recall.

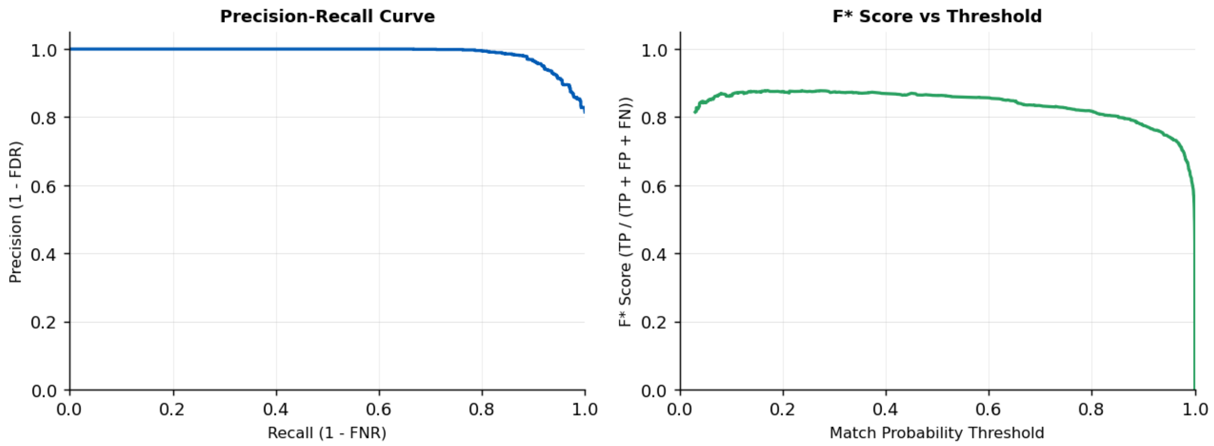


*Figure: Precision-Recall curve (left) and F* vs threshold (right).*

## CRL Score (Composite Reliability of Linkage)

The CRL score measures linkage quality in the valid operating region where both FDR (False Discovery Rate) and FNR (False Negative Rate) are within the acceptable epsilon threshold. CRL = AVG(F*) * (t_upper - t_lower). A higher CRL score indicates more reliable linkage.

| | |
|---|---|
| **Epsilon (tolerance):** | 0.10 |
| **CRL Score:** | 0.211434 |
| **t_upper:** | 0.3441226639519589 |
| **t_lower:** | 0.10221102897486595 |
| **epsilon_z:** | 0.07070707070707072 |